

Foundations of All Concepts Implemented in This Project

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1 Abstract

This project will cover several GAN

2 Introduction

In 2014 a certain type of generative models was invented by Goodfellow et al. [1], these new type of neural networks are called *Generative Adversarial Networks* (GANs). Compared to new generative models like **DALL-E** developed by Open-AI, which is based on *Diffusion Models* (These models will probably come also in a own project as soon as GANs are accomplished), their ability of creating diverse and realistic looking images is limited. This comes with the cost of relevance in modern applications. However, GANs are still used in a wide range of applications from data augmentation and self supervised learning. One of the biggest advantage still remains in the fast inference compared to the diffusion models.

2.1 GAN Issues

Training GANs is inherently complex firstly due to their non convergence nature. This is easily seen by equation 1, their objective is against each other. Therefore the training is instable and may not converge til a global minima.

Secondly, due to their ability to be stuck in a local minima and only produce limited modes of the actual training data. The generator is only producing partiell parts of the training data. For instance, imagine that the generator is only producing the number 6 and 7 but not the others. Unfortunately, if the discriminator is not well trained only some nodes are sufficiently to fool the discriminator. This is called Mode colapse.

3 Architectures

3.1 Vanilla GAN

The beginning of this hopefully reasonably overview, will be the start of GANs with Goodfellow et al. [1], which might be the most influential paper about GANs. Shortly said, GANs consist of two neural networks working against each other to surpass the performance of the counterpart. One of the participants is the Generator that tries to generate data that resamples the original data. This procedure consists on the assumption that each dataset can be seen as a probability distribution. Following, this assumption the generator tries to learn the distribution of the dataset and can sample realistic looking images from the learned distribution. The Discriminators' objective is to distinguish whether real or generated data (fake data) was fed into the discriminator. The Generator learns to fool the discriminator, while the discriminator has to learn to distinguish. Mathematically, this can be seen with expression:

$$\min_G \max_D V(D, G) = \mathbb{E}_{x \sim p_{\text{data}}(x)} [\log D(x)] + \mathbb{E}_{z \sim p_z(z)} [\log (1 - D(G(z)))] . \quad (1)$$

the objective of the discriminator is actually to maximize the probability of whether beeing real in the first term or zero in the second term.

3.2 DCGAN

3.3 Resnet-GAN

3.4 Conditional GAN

3.5 Super Resolution GAN

3.6 BigGAN

4 Objective Functions

4.1 Least Square Loss

4.2 Wasserstein GAN

4.3 Wasserstein GAN with Gradient Penalty

4.4 Hinge GAN

5 Regularizers

5.1 Feature Matching

5.2 Minibatch-Discrimination

5.3 Spectral Normalization

5.4 Orthogonal Normalization

6 Experiments

6.1 Methods

7 Combinations tried

An exemplary overview of the structure that will follow soon, some trainings are in progress.

Table 1: Overview of the evaluated GAN configurations.

ID	Architecture	Loss	Regularization
A	DCGAN	Vanilla GAN	None
B	DCGAN	LSGAN	Spectral Normalization
C	DCGAN	LSGAN + Feature Matching	Spectral Normalization
D	DCGAN	LSGAN	Minibatch Discrimination
E	DCGAN	LSGAN	Label Smoothing
F	DCGAN	LSGAN	Spectral Normalization + Label Smoothing
G	DCGAN	LSGAN	Spectral Normalization + Minibatch Discrimination
H	DCGAN	WGAN	Weight Clipping
I	DCGAN	WGAN-GP	Gradient Penalty

8 Summary

9 Appendix

References

- [1] Ian Goodfellow et al. “Generative Adversarial Nets”. In: *Advances in Neural Information Processing Systems*. 2014.